



Artificial neuron that mimics nerve cells can be used for medical treatments

An artificial organic neuron that closely mimics the characteristics of biological nerve cells has been created by researchers at Linköping University (LiU), Sweden. This



Padinhare Cholakkal Harikesh, lead author of the new study, makes the chemical transistors used in the artificial neurons (Credit: Thor Balkhed)

artificial neuron can stimulate natural nerves, making it a promising technology for various medical treatments in the future. This new artificial nerve cell called “conductance based organic electrochemical neuron” or c-OECN, closely mimics 15 out of the 20 neural features that characterise biological nerve cells, making its functioning much more similar to natural nerve cells. The device uses ions that control the flow of electric current through an n-type conducting polymer, leading to spikes in the device’s voltage. The artificial neuron can stimulate the vagus nerve itself and could pave the way for essential applications in various forms of medical treatment.

A new system barely uses any power for short-range communication

Researchers have developed a communication system that barely uses any power to operate for short-range communication applications. This new system enables wireless transmission of data simply by opening and closing a switch that connects a resistor to an antenna; no need to send power to the antenna. When combined with techniques for harvesting energy from the environment, it could lead to all manner of devices that transmit data, including tiny sensors and implanted medical devices, without needing batteries or other power sources. Apart from the energy needed to flip the switch, no other energy is needed to transmit the information. In our case, the switch is a transistor, an electrically controlled switch with no moving parts that consumes a minuscule amount of

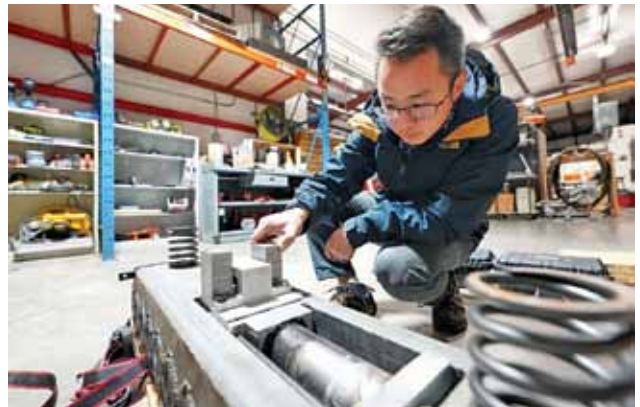


This experimental setup shows an ultra-low-power wireless communications device that could one day be used in tiny remote sensors (Credit: Zerina Kapetanovic, CC BY-ND)

power. A switch connects and disconnects a strong electrical signal source—perhaps an oscillator that produces a sine wave fluctuating two billion times per second—to the transmit antenna.

Harvesting electromagnetic energy from moving trains made possible

Researchers from the Virginia Tech Center for Vehicle Systems and Safety (CVeSS) and the Railway Technologies Laboratory have found a way to harvest electromagnetic energy from rail transportations. They have created a new



Yang Chen inspects an energy-harvesting rail in the Railway Technologies Laboratory (Credit: Alex Parrish for Virginia Tech)

kind of tie that replaces the conventional wooden variety and is equipped to generate power. This tie is placed underneath the rail, topped with a heavy metal bar mounted on a spring. As the wheels of the train pass over the rail, the train’s weight pushes down on that bar, triggering a series of gears. Those gears rotate a generator, creating electricity, which can then be stored in a battery. Researchers harvested 15 to 20 watts of power for every wheel of the train passing over the rail.